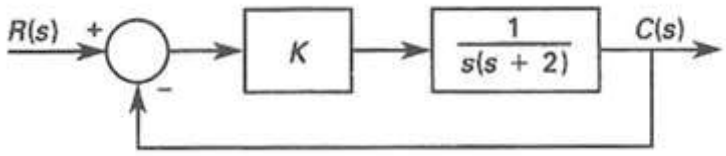
B - Mechanical Design MS4092

* Wajib

B - Mechanical Design MS4092

Number of problems: 8
Start at: 10.00 AM
Finish at: 10.24 AM
Duration: 24 Minutes

For the given block diagram, K is set to 25. The damped natural frequency of oscillation in response to a step input is most nearly... *



unit impulse : $F(s)=1$

step unit : $F(s)=\frac{1}{s}$

- 4.9
- 3.9
- 5.9
- 2.9

$$\frac{\frac{k}{s(s+2)}}{1+\frac{k}{s(s+2)}} = \frac{k}{s^2+2s+k}$$

$\omega_n = 5$

$2\zeta\omega_n = 2$

$\zeta = 0,2$

$\omega_d = 5\sqrt{1-0.2^2}$

$\omega_d = 4,9$

The rotor of a steam turbine is rotating at 3600 rpm when the steam supply is suddenly cut off. The rotor decelerates at a constant rate and comes to rest after 5 min. What is most nearly the angular deceleration of the rotor in rad/s2?

- 0.40
- 0.85
- 1.25
- 2.50

$$\omega = \frac{3600 \times 2\pi}{60} = 120\pi \text{ rad/s}$$

$$\alpha = \frac{120\pi}{5 \times 60} = 1,25 \text{ rad/s}^2$$

An 18-tooth straight spur gear transmit s a torque of 1200 N·mm. The pitch circle diameter is 20 mm, and the pressure angle is 14.5°. What is most nearly the radial force on the gear?

- 31 N
- 17 N
- 109 N
- 120 N

$T = 1200$ Nmm $r = 10$ mm

$\phi = 14,5°$

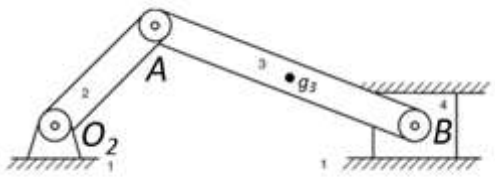
$$F_t = \frac{T}{r} = 120 \text{ N}$$

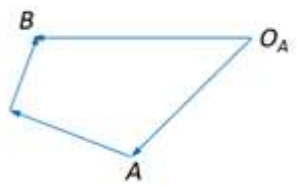
$$F_r = F_t \times \tan\phi$$

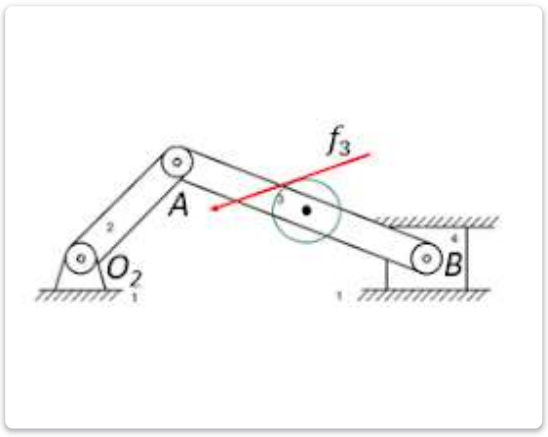
$$= 120 \times \tan 14.5°$$

$F_r = 31$ N

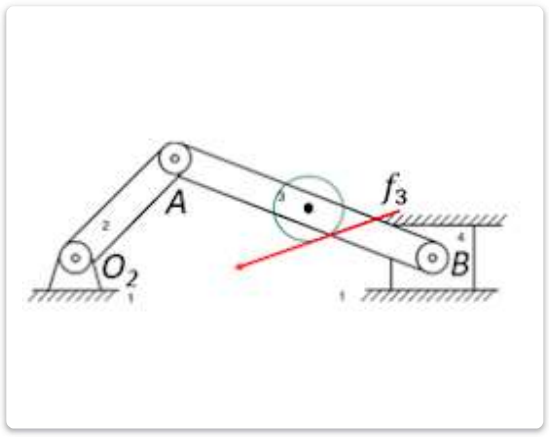
In the slide-crank mechanism and its acceleration polygon below, the correct inertia force of link 3 is...



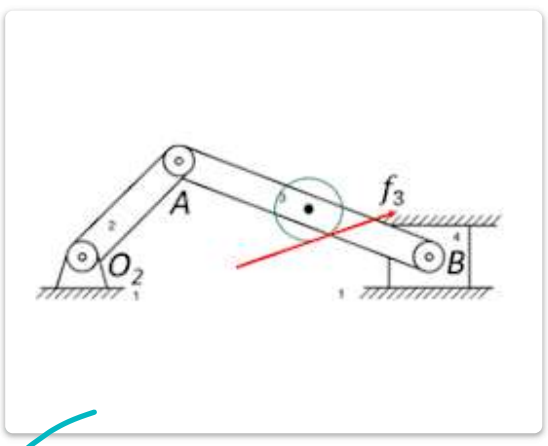




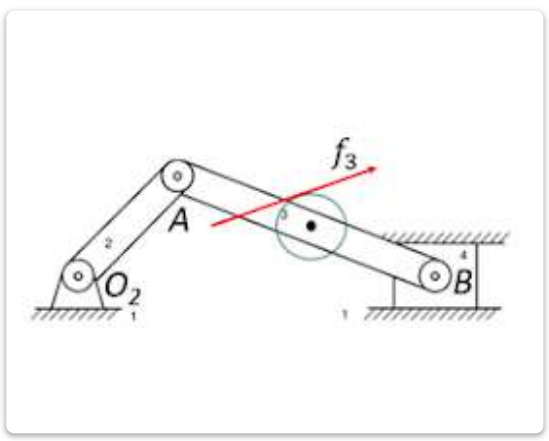
Option 2



Option 4

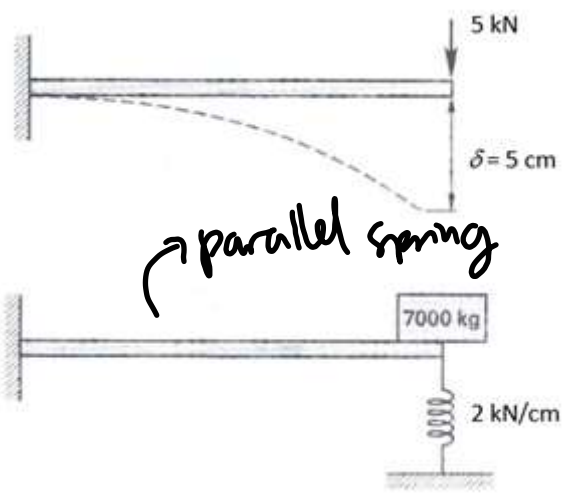


Option 3



Option 1

A cantilever beam deflects 5 cm when a force of 5 kN is applied at the end. The beam tip is subsequently supported by a spring with a stiffness of 2 kN/cm. When a 7000 kg mass is attached at the beam's tip, what is most nearly the natural frequency of the massbeam-spring system?



$k_1 = \frac{P}{\delta_{max}} = \frac{5\ kN}{5\ cm} = 1\ kN/cm$

$k_2 = 2\ kN/cm$

$\omega_n = \sqrt{\frac{k_1 + k_2}{m}}$

$= \sqrt{\frac{3000 \times 100}{7000}}$

$\omega_n = 6.5\ rad/s$

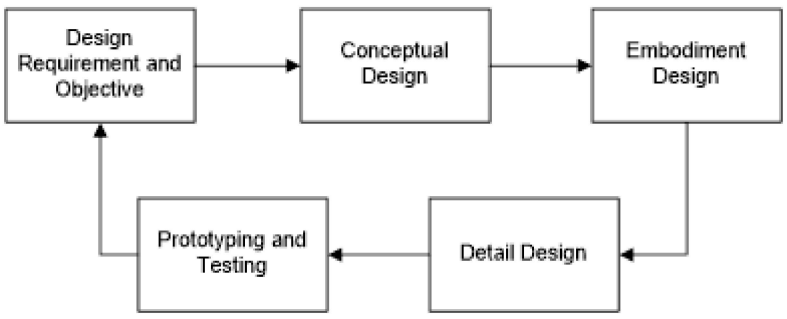
7.0 rad/s

6.0 rad/s

7.5 rad/s

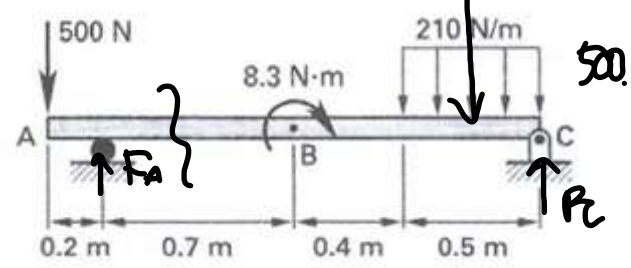
6.5 rad/s

Figure below shows the engineering design steps in general. Which statement below is false?



- The parametric design is the result of detail design.
- Analytic Hierarchy Process (AHP) in evaluation is the part of conceptual design step.
- There should be background research in the identification of design requirement and objective. ✓
- Configuration design generation is included in the embodiment design.

The overhanging beam shown is supported by a roller and a pinned support. What is the internal bending moment of point D that is located 0.6 m from point A? *



$52,5\ N$

$F_A + F_C = 552,5\ N$

$500.1,8 + 52,5 . 0,25 = F_A . 1,6 + 8,3$

$F_A = 558,13\ N$

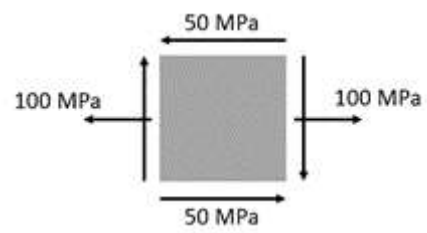
$F_C = -5,63\ N$

- -62 Nm
- -70 Nm
- -86 Nm
- ✗ -78 Nm

$M_x = 558,13(x - 0,2) - 500x$

$M_x = -76,75\ Nm$

A plane element in a body is subjected to the stresses shown. What is most nearly the maximum shear stress?



- 96 MPa
- 121 MPa
- ✗ 71 MPa
- 46 MPa

$\sigma_x = 100\ MPa$

$\sigma_y = 0$

$\tau_{xy} = 50\ MPa$

$\sigma_{avg} = \frac{\sigma_x + \sigma_y}{2}$

$\tau_{xy\,max} = R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

Kembali Kirim

Jangan pernah mengirimkan sandi melalui Google Formulir.

Konten ini tidak dibuat atau didukung oleh Google. Laporkan Penyalahgunaan - Persyaratan Layanan - Kebijakan Privasi

Google Formulir

$$\tau_{xy\,max} = \sqrt{\left(\frac{100-0}{2}\right)^2 + 50^2}$$

$$= 70{,}71 \text{ MPa} \approx 71 \text{ MPa}$$